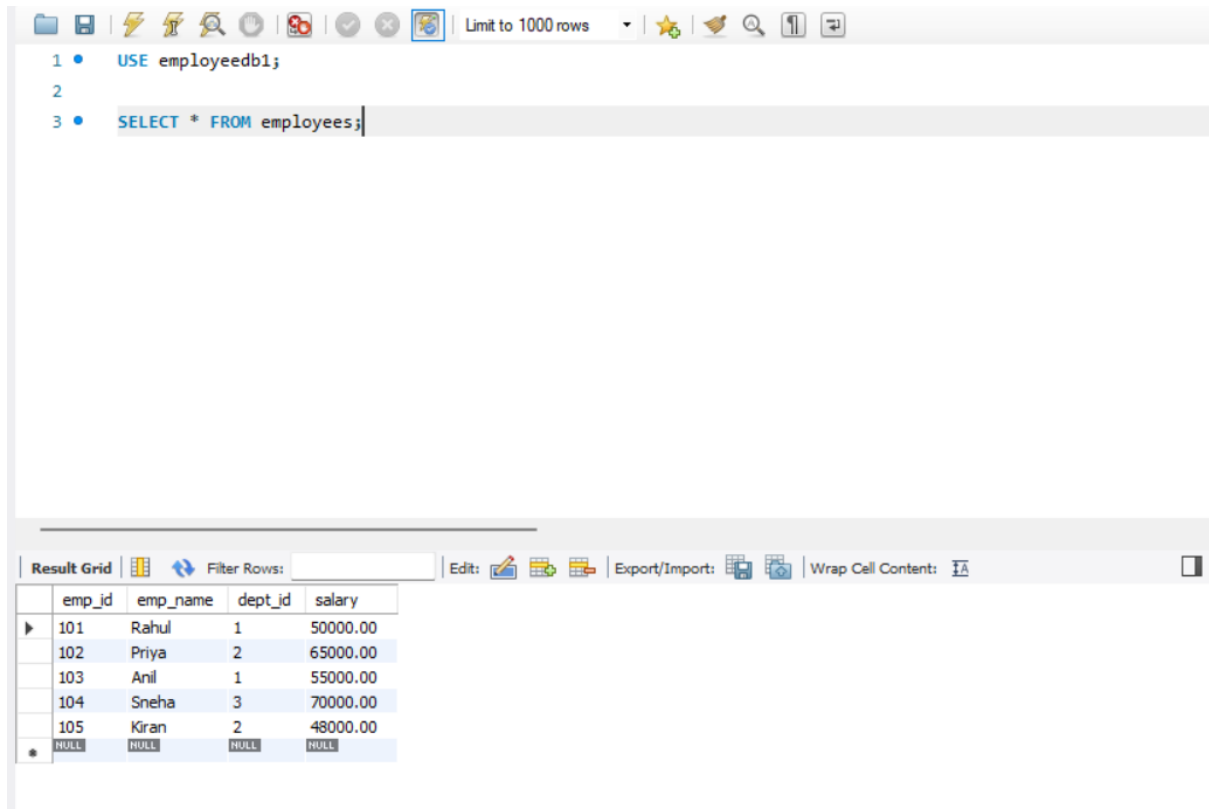


DBMS – MYSQL STORED PROCEDURES

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Employee Database and Table Data



The screenshot displays a MySQL IDE interface. The top toolbar includes icons for file operations, execution, and search. The SQL editor contains the following code:

```
1 • USE employeeedb1;  
2  
3 • SELECT * FROM employees;
```

Below the editor, the 'Result Grid' tab is active, showing the output of the query. The grid has columns for emp_id, emp_name, dept_id, and salary. The data is as follows:

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00
* NULL	NULL	NULL	NULL

Procedure to Display All Employees

Limit to 1000 rows

```
1 • USE employeeedb1;
2
3 • SELECT * FROM employees;
4 • DROP PROCEDURE IF EXISTS GetEmployees;
5
6 DELIMITER $$
7
8 • CREATE PROCEDURE GetEmployees()
9 BEGIN
10     SELECT * FROM employees;
11 END $$
12
13 DELIMITER ;
14 • CALL GetEmployees();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

Procedure Using IN Parameter

```
16 • USE employeeedb1;
17
18 • DROP PROCEDURE IF EXISTS GetEmployeeById;
19
20 DELIMITER $$
21
22 • CREATE PROCEDURE GetEmployeeById(IN eid INT)
23 BEGIN
24     SELECT *
25     FROM employees
26     WHERE emp_id = eid;
27 END $$
28
29 DELIMITER ;
30 • CALL GetEmployeeById(102);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

emp_id	emp_name	dept_id	salary
102	Priya	2	65000.00

Procedure to Increase Employee Salary

The screenshot shows a MySQL IDE window with a SQL script editor and a result grid. The script defines a procedure `IncreaseSalary` that takes an employee ID and an amount, and updates the employee's salary. It then calls this procedure for employee 101 with an amount of 5000, followed by a select statement to view the updated data.

```
36 DELIMITER $$
37
38 CREATE PROCEDURE IncreaseSalary(
39     IN eid INT,
40     IN amount DECIMAL(10,2)
41 )
42 BEGIN
43     UPDATE employees
44     SET salary = salary + amount
45     WHERE emp_id = eid;
46 END $$
47
48 DELIMITER ;
49 CALL IncreaseSalary(101, 5000);
50 SELECT * FROM employees
51 WHERE emp_id = 101;
```

The result grid below the script shows the data for employee 101 after the procedure execution:

emp_id	emp_name	dept_id	salary
101	Rahul	1	60000.00
NULL	NULL	NULL	NULL

Procedure Using Aggregate Function

The screenshot shows a MySQL IDE window with a SQL script editor and a result grid. The script drops an existing procedure `AvgSalary`, creates a new one that calculates the average salary of all employees, and then calls the procedure.

```
52
53
54 USE employeeedb1;
55
56 DROP PROCEDURE IF EXISTS AvgSalary;
57
58 DELIMITER $$
59
60 CREATE PROCEDURE AvgSalary()
61 BEGIN
62     SELECT AVG(salary) AS AverageSalary
63     FROM employees;
64 END $$
65
66 DELIMITER ;
67 CALL AvgSalary();
```

The result grid below the script shows the output of the `AvgSalary` procedure:

AverageSalary
61600.000000

Procedure Using JOIN

Limit to 1000 rows

```

97
98     DELIMITER $$
99
100 • CREATE PROCEDURE DepartmentAverageSalary()
101     BEGIN
102         SELECT d.dept_name,
103                COUNT(e.emp_id) AS EmployeeCount,
104                AVG(e.salary) AS AverageSalary
105         FROM employees e
106         JOIN departments d
107           ON e.dept_id = d.dept_id
108         GROUP BY d.dept_id, d.dept_name;
109     END $$
110
111     DELIMITER ;
112 • CALL DepartmentAverageSalary();

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [A](#)

	dept_name	EmployeeCount	AverageSalary
▶	CSE	2	77500.000000
	ECE	2	56500.000000
	EEE	1	70000.000000

IF...ELSE Salary Classification

```

121 • BEGIN
122     DECLARE sal DECIMAL(10,2);
123
124     SELECT salary INTO sal
125     FROM employees
126     WHERE emp_id = eid;
127
128     IF sal >= 60000 THEN
129         SELECT 'High Salary' AS Status;
130     ELSE
131         SELECT 'Low Salary' AS Status;
132     END IF;
133 END $$
134
135     DELIMITER ;
136 • CALL SalaryStatus(102);

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [A](#)

	Status
▶	High Salary

OUT Parameter for Total Employee Count

```

151 • CALL EmployeeCount(@total);
152 • SELECT @total AS TotalEmployees;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
TotalEmployees			
5			

INOUT Parameter

```

167 • CALL AddBonus(@bonus);
168
169 • SELECT @bonus AS UpdatedBonus;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
UpdatedBonus			
15000.00			

Transaction

```

188 UPDATE employees
189 SET salary = salary + amount
190 WHERE emp_id = toEmp;
191
192 COMMIT;
193 END $$
194
195 DELIMITER ;
196 • SELECT *
197 FROM employees
198 WHERE emp_id IN (101,102);
199 • CALL TransferSalaryAmount(101,102,2000);
200
201 • SELECT *
202 FROM employees
203 WHERE emp_id IN (101,102);

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
emp_id	emp_name	dept_id	salary	
101	Rahul	1	151000.00	
102	Priya	2	69000.00	
*	NULL	NULL	NULL	

STUDENT PRACTICE QUESTIONS

Employees Above Given Salary

```
207 • DROP PROCEDURE IF EXISTS EmployeesAboveSalary;
208
209 DELIMITER $$
210
211 • CREATE PROCEDURE EmployeesAboveSalary(IN min_salary DECIMAL(10,2))
212 BEGIN
213     SELECT *
214     FROM employees
215     WHERE salary > min_salary;
216 END $$
217
218 DELIMITER ;
219 • CALL EmployeesAboveSalary(50000);
220
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

emp_id	emp_name	dept_id	salary
101	Rahul	1	157000.00
102	Priya	2	73000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00

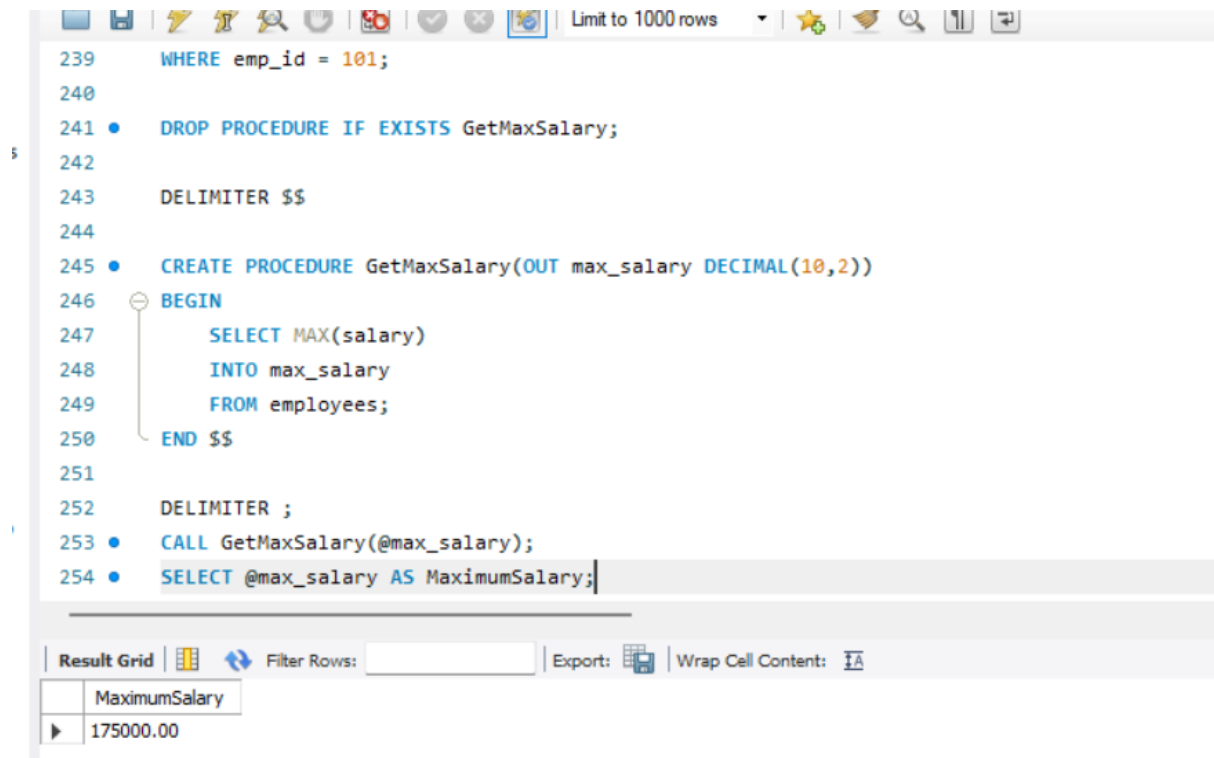
Update Employee Department

```
224
225 • CREATE PROCEDURE UpdateEmployeeDepartment(
226     IN employee_id INT,
227     IN new_department_id INT
228 )
229 BEGIN
230     UPDATE employees
231     SET dept_id = new_department_id
232     WHERE emp_id = employee_id;
233 END $$
234
235 DELIMITER ;
236 • CALL UpdateEmployeeDepartment(101, 2);
237 • SELECT *
238 FROM employees
239 WHERE emp_id = 101;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

emp_id	emp_name	dept_id	salary
101	Rahul	2	166000.00
*	NULL	NULL	NULL

Maximum Salary Using OUT

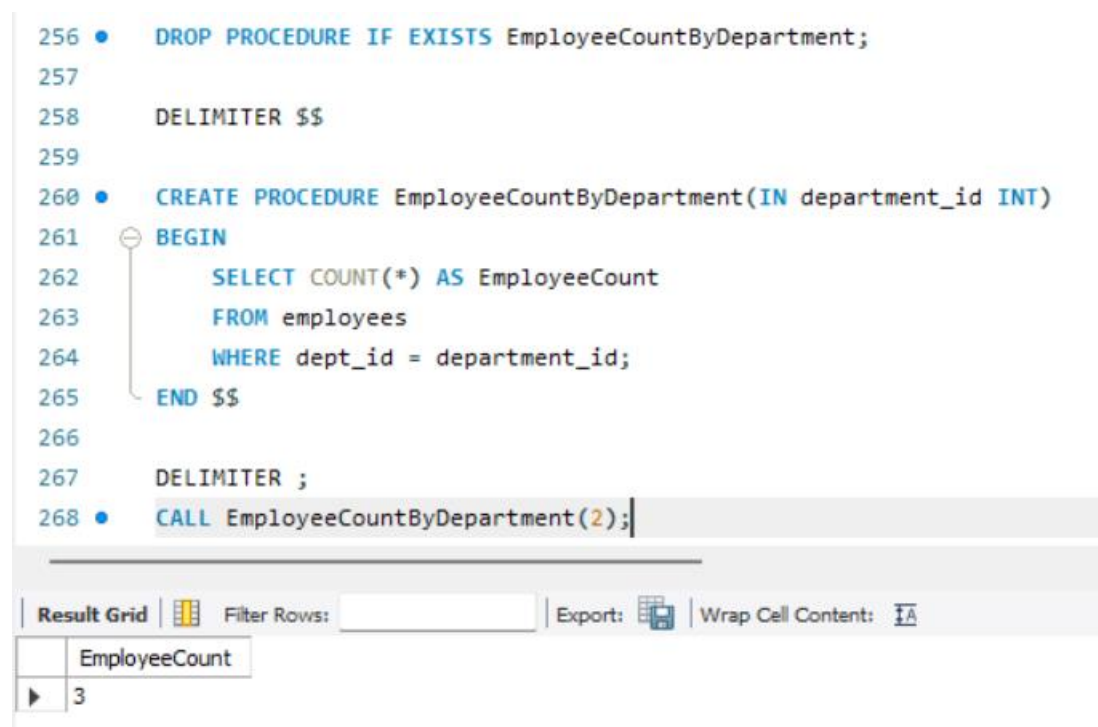


```
239 WHERE emp_id = 101;
240
241 • DROP PROCEDURE IF EXISTS GetMaxSalary;
242
243 DELIMITER $$
244
245 • CREATE PROCEDURE GetMaxSalary(OUT max_salary DECIMAL(10,2))
246 BEGIN
247     SELECT MAX(salary)
248     INTO max_salary
249     FROM employees;
250 END $$
251
252 DELIMITER ;
253 • CALL GetMaxSalary(@max_salary);
254 • SELECT @max_salary AS MaximumSalary;
```

Result Grid

MaximumSalary
175000.00

Employee Count of a Department



```
256 • DROP PROCEDURE IF EXISTS EmployeeCountByDepartment;
257
258 DELIMITER $$
259
260 • CREATE PROCEDURE EmployeeCountByDepartment(IN department_id INT)
261 BEGIN
262     SELECT COUNT(*) AS EmployeeCount
263     FROM employees
264     WHERE dept_id = department_id;
265 END $$
266
267 DELIMITER ;
268 • CALL EmployeeCountByDepartment(2);
```

Result Grid

EmployeeCount
3

JOIN Employee + Department Details

```

275 • CREATE PROCEDURE EmployeeDepartmentDetails()
276 BEGIN
277     SELECT
278         e.emp_id,
279         e.emp_name,
280         e.salary,
281         d.dept_id,
282         d.dept_name
283     FROM employees e
284     JOIN departments d
285     ON e.dept_id = d.dept_id;
286 END $$
287
288 DELIMITER ;
289 • CALL EmployeeDepartmentDetails();

```

emp_id	emp_name	salary	dept_id	dept_name
101	Rahul	187000.00	2	ECE
102	Priya	93000.00	2	ECE
103	Anil	55000.00	1	CSE
104	Sneha	70000.00	3	EEE
105	Kiran	48000.00	2	ECE

Departments With Average Salary > 55000

```

306 • CALL DepartmentsHighAverageSalary();

```

dept_id	AverageSalary
2	112666.666667
3	70000.000000

IF...ELSE Salary Classification

```

332 • CALL ClassifySalary(101);

```

Salary	SalaryLevel
199000.00	High

Average Salary of Department

```

333
334 • DROP PROCEDURE IF EXISTS DepartmentAverageSalary;
335
336 DELIMITER $$
337
338 • CREATE PROCEDURE DepartmentAverageSalary(IN department_id INT)
339 BEGIN
340     SELECT AVG(salary) AS AverageSalary
341     FROM employees
342     WHERE dept_id = department_id;
343 END $$
344
345 DELIMITER ;
346 • CALL DepartmentAverageSalary(2);

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

AverageSalary
119333.333333

INOUT Bonus

```

360 • SET @salary = 50000;
361 • CALL ApplyBonus(@salary);
362 • SELECT @salary AS SalaryAfterBonus;
363

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

SalaryAfterBonus
55000.00

Transaction With Two Updates

```

392 • CALL TransferSalary(101, 102, 2000);
393 • SELECT *
394 FROM employees
395 WHERE emp_id IN (101, 102);

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

emp_id	emp_name	dept_id	salary
101	Rahul	2	225000.00
102	Priya	2	125000.00
NULL	NULL	NULL	NULL